

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندوی	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for: Industry partner, investor or funder

Contents

Part I: Orientation 2

- 2. The work, not the worker 3
- 3. Start here: find your way in 4

Part II: One architecture 6

- 6. Architecture: core, projections, layers 7

Part V: Working with the estate 12

- 29. The roadmap, for every collaborator and every domain 13
- 30. What the work would cost: a model-based estimate 20

Part VI: Status 22

- 31. This release and the next 23

Appendices 25

- E. For practitioners: teachers, policy makers, communities, makers and engineers 26

Part I: Orientation

2. The work, not the worker

In plain words. What matters is whether the work holds up, whether the code runs and the reasoning is sound, not who made it. The aim is that people build technology in their own languages.

We know Pāṇini almost entirely through his grammar. Tradition gives a birthplace, Śalātura in Gandhāra, and very little else that can be checked. Close to four thousand sūtras survive; the person does not. The grammar still works, and Chapter 20 runs part of it. That is the standard this guide keeps. Work has to stand without its author, because titles, affiliations and biographies do not compile, and code, derivations and measurements do.

Two consequences follow for how the estate is built.

The first is enablement. Linguistic and cultural equity is not achieved by making AI available everywhere. Systems have to work within the languages, scripts, cultural contexts and knowledge traditions of the people they serve, so that those people become producers of technology in their own languages rather than remaining users of someone else's.

The second is openness. Every repository in Chapter 12 can be read, run, forked and criticised. Nothing here needs anyone's permission to use; each repository states its licence.

The programme stands on two pillars. The linguistic pillar standardises at the level of the dialect, with the idiolect as a later overlay, and grows through linguists working across the world. The technological pillar, the older of the two, is now the enabler: the transliteration engines, compilers, parsers and substrates that make a language usable for building. This guide deals mostly with the second pillar, because that is where a newcomer can start contributing today.

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part II: One architecture

6. Architecture: core, projections, layers

In plain words. One shared core, Humanesque, is seen through three cultures: Indian, Persian and Western. Around it sit the parts that turn an idea, in any language, into something real.

One core, three projections

Humanesque is the core: one shared, multidimensional, executable typed hypergraph, with a federated ontology of 26 positions, O_0 to O_{25} , a common execution and evidence spine, and components that can be implemented independently. Its GitHub organisation is hmnsq, where every core stem repository is to live.

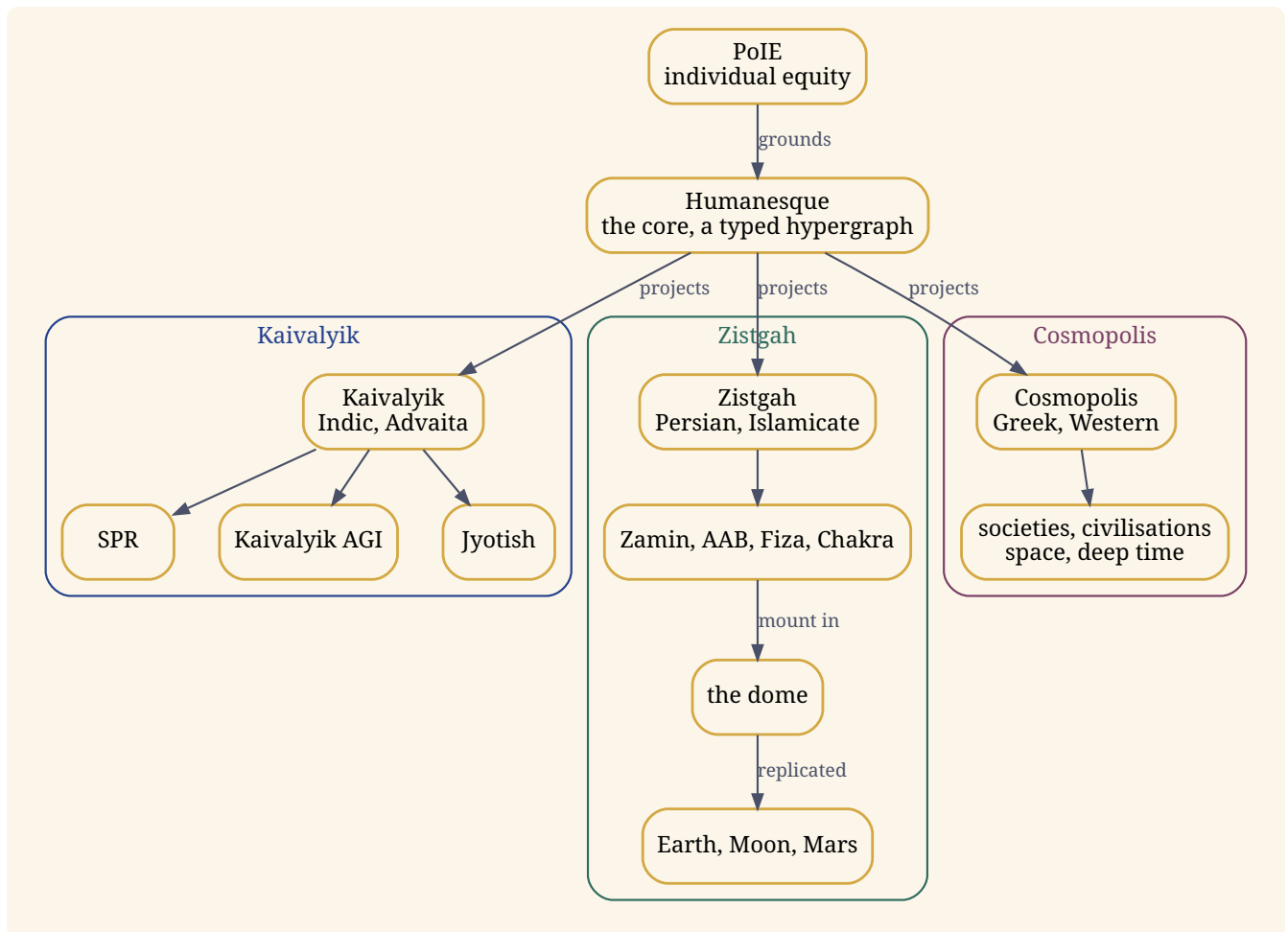


Figure 2. Humanesque and its projections: parallel cultural traversals, not stages.

Kaivalyik, Zistgah and Cosmopolis are three projections over it, running in parallel: the Indic, including the Advaita framing; the Persian and Islamicate; and the Greek, Western and classical. A projection is a domain-specific view and traversal of the whole architecture, with its own culturally sensitive naming and branding. The projections are not stages, not levels of a hierarchy and not versions of one another; they intersect, diverge and reconverge, and none completes another. Zistgah is not the parent of the other two. Their organisations are kaivalyik, zistgah and c-polis; today zistgah carries the working repositories.

The PANINI stack

Layer	What it holds	Where to look
Front end	The languages a program or a prompt can arrive in, Sanskrit first: ILM and the language parsers	project-ilm/ilm.codes, zistgah/tajziya
Middleware	PANINI's own front ends, from <code>ada.pni</code> to <code>zig.pni</code> and <code>sysml.pni</code> , and the PANINI language itself	zistgah/panini, zistgah/humanesque
Core PANINI	The construct model: common semantics plus decorators	zistgah/humanesque, <code>ilm/constructs.csv</code> and <code>ilm/decorators.csv</code>
Realization backends	Where a program meets a substrate: host toolchains for code; PANINIq for oscillator and quantum substrates; PANINIB for biology; PANINIphy for physical systems	zistgah/paniniq, zistgah/paninib, zistgah/paniniPHY

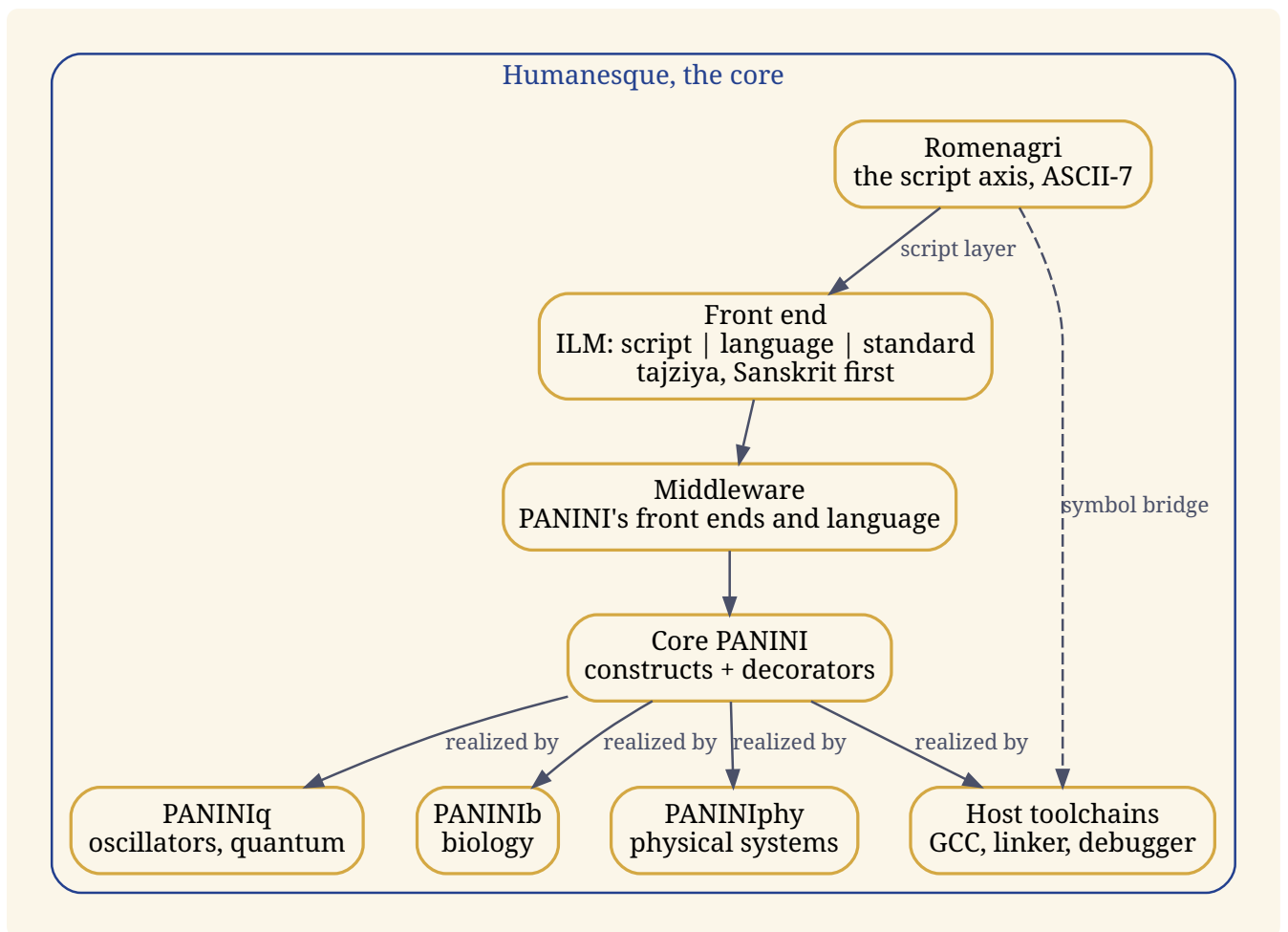


Figure 3. The PANINI stack, with Romenagri as the script layer beneath every front end.

Beneath every front end lies the script layer, Romenagri. Every keyword and every name of a program becomes a word over A to Z, a to z and the underscore, reversibly, so that the whole existing toolchain, from lexer to linker to debugger, carries it unchanged, and the inverse renders it back in the script wherever a person reads it. Chapters 18, 19 and 21 explain it and run it.

A construct such as a counted loop is defined once in the core and translated once per human language. A decorator carries what a particular host language drags in with it, such as `stdio.h` for printing in C. Core PANINI, in his words, “does not violate anything”; the front ends and the realization backends are all projections of it.

How the parts compose

Humanesque	the core; Kaivalyik, Zistgah and Cosmopolis project it in parallel; everything below lives in it, nothing below leads to it
front end	ILM (script language standard) and tajziya, Sanskrit first the script axis is Romenagri, which persists as the symbol bridge through compiler, linker, ELF, DWARF and debugger
middleware	PANINI's own front ends and the PANINI language
core	core PANINI: construct model = common semantics + decorators
realization	host toolchains for code; PANINIq (oscillator, quantum); PANINIb (biology, down to sequence IR); PANINIphy (physical systems); PRATIK and Zamin as a balanced-ternary substrate

Around the stack: the cyclers are written in PANINI and run by one engine; GENIE composes and runs research cycles; AAB paints tasks and runs the process cyclers; FAKIR, UKOP and Dhancha supply the domain lattice and its invariants; Mez, the Cognitive Workbench, composes any of them for an exercise without absorbing them; the provenance systems seal every artifact; VGC gates the work between people and AI. Two readings must be kept apart. As genealogy, the order is Romenagri, Hindawi, ILM, PANINI and the realization arms. As architecture, Romenagri sits inside ILM's script axis, the cyclers are written in PANINI rather than downstream of it, and Humanesque is the core that contains the whole, not its end point.

The realization spine

The realization repositories state the spine they share. PANINI lowers declarative intent through language, semantic IR, a realization requirement, domain resolution and a domain IR, and then a common layer of geometry, trajectory, animation, observation, verification and an ArtifactGraph. PANINIb lowers biological intent through semantic, resolution, CISC, RISC and sequence IRs and emits two coupled artifacts, a physical realization plan and a digital execution run; PANINIphy lowers physical intent through a physical IR and physical-domain engines. “IR contracts are the product. Laboratories and machines are external adapters.” And the same repositories keep the states of a result apart: desired, predicted, observed and validated phenotype are four different things, as are compilation success, execution success, target satisfaction and biological validation.

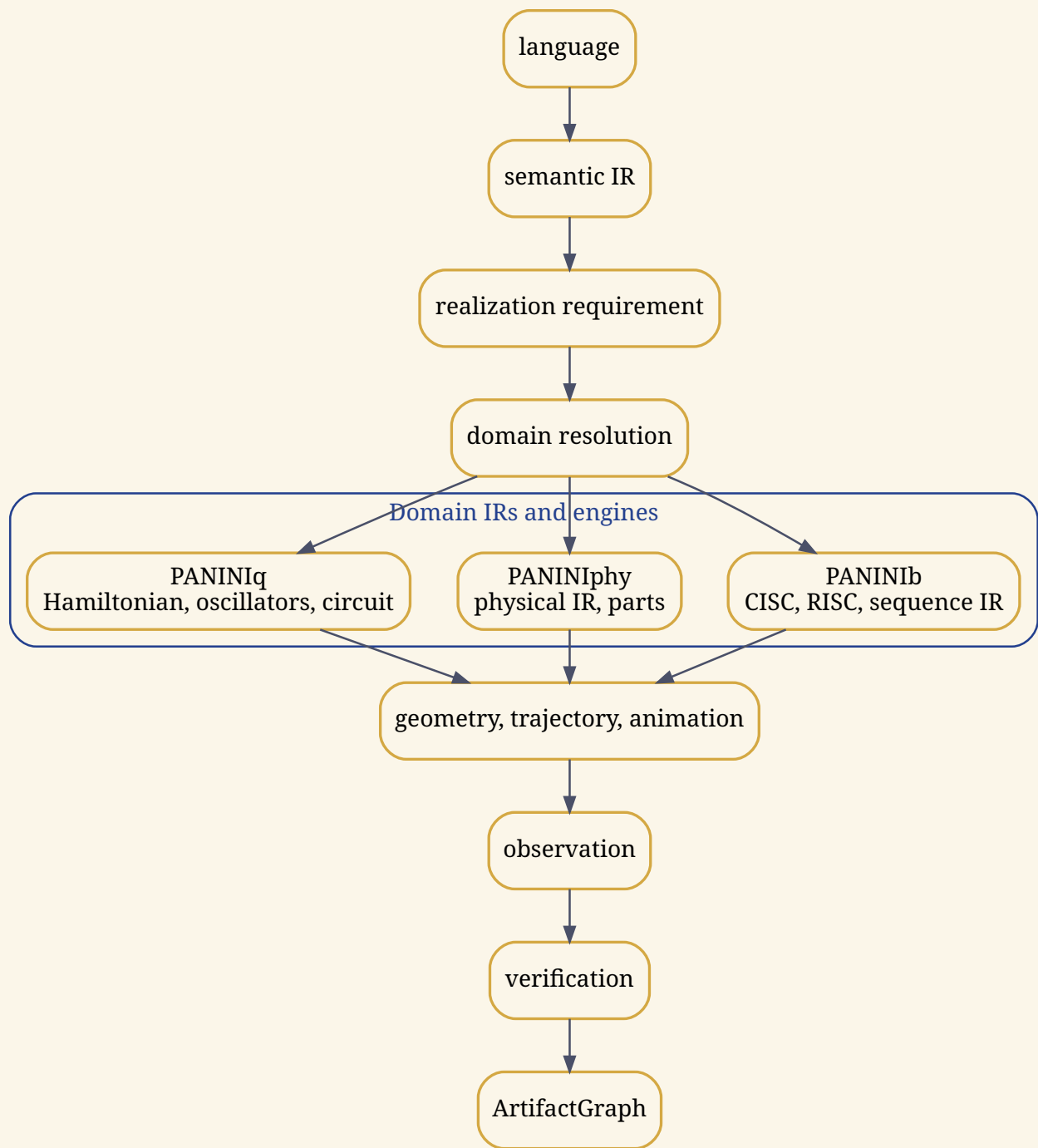


Figure 4. The realization spine the realization arms share.

The habitat elements

Zamin is the ground: the balanced-ternary hardware of PRATIK. AAB is the water: the paint program for systems, from painted intent to verified, sealed artifact, where the verification-gated quests run. Fiza is the air: the environmental replica. Chakra is the turning sky: the observatory and its calendars. All of them mount in one shared virtual dome, which the habitats on the Moon and Mars replicate.

The cyclers

A cycler is a sequence of prompts and the outputs they produce, written down so that it can be edited, shared and run with any AI. Six are classified by what they produce: matba (print), khwab (visual), awaz (audio), tilasm (immersive), pench (embodied) and yadein (the record); genie runs research cycles. Each has its own purpose, contract, context, state model, invariants, failure modes, evidence requirements, workflow and artifact model; only the engine is shared. The protocol behind every cycle, in the author's words: intent, then context, then a meaningful prompt; any AI answers and the human inspects; the artifacts and responses are kept under configuration management; the next prompt follows, until a final artifact that the human authored with intention.

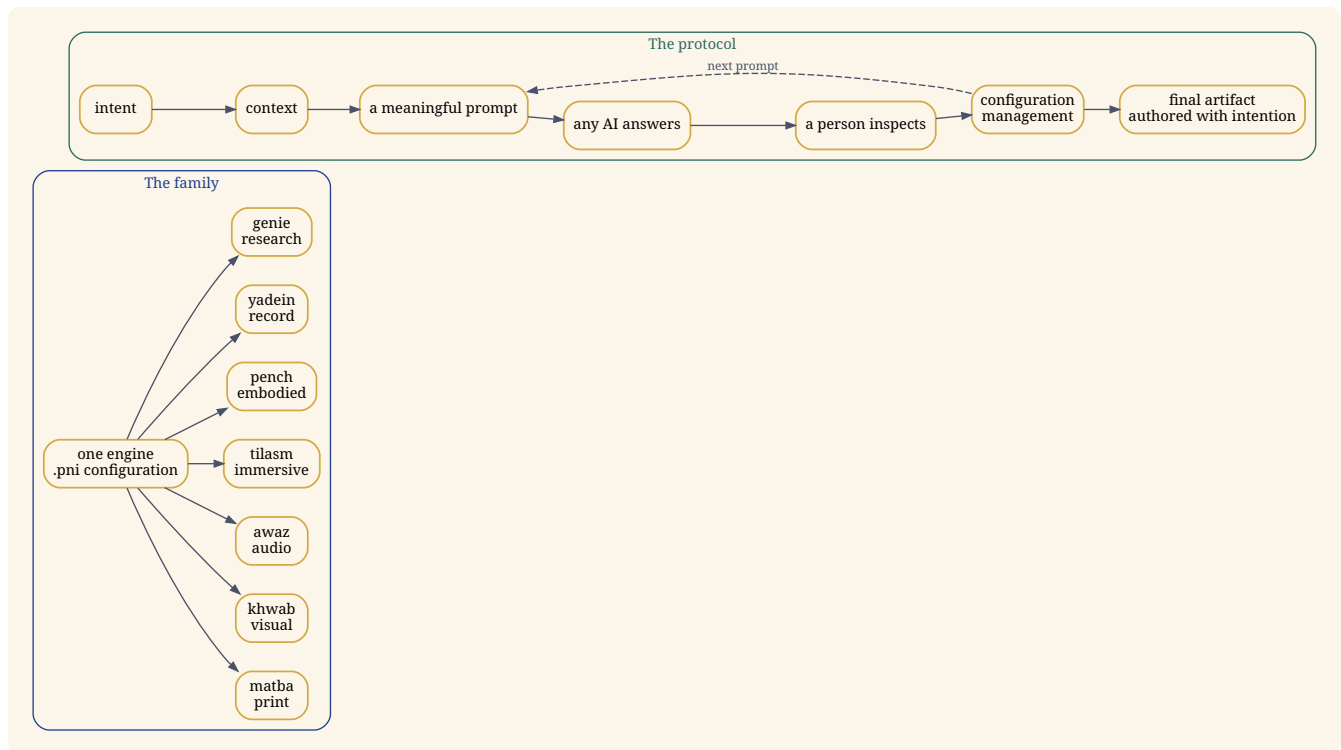


Figure 5. The cyclers: one engine, configured in PANINI, and the protocol every cycle follows.

Provenance, in a fixed order

Four systems, never collapsed into one. **Tok DOI** proves that a thing existed by a given time. **spiguard** is the disclosure gate, which fails closed. **Candor** records signed intent as in-toto statements. **Misty DOI** is the one that reaches the world, through Zenodo. The order is always seal, clear, attest, mint. Chapter 25 explains the machinery.

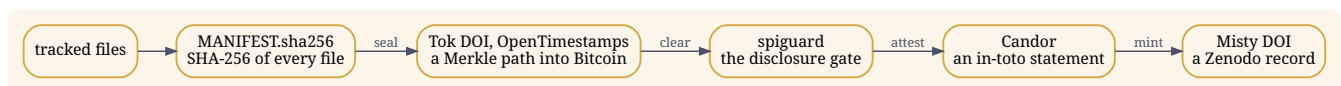


Figure 6. Provenance in its fixed order: seal, clear, attest, mint.

Part V: Working with the estate

29. The roadmap, for every collaborator and every domain

In plain words. The plan to finish the releases and open the work to everyone: what happens when, which lane each kind of person can join, and every field of work, from farming to medicine to software.

The plan to complete the releases and to open the estate to collaborators is kept as data and run as a GitHub project: one board, an epic for each phase, and an issue for every piece of work, each with its finding, its task, the exact gate that proves it done, the evidence to attach, and what it depends on. The same data gives every collaborator a lane and every domain a lane, so anyone can find where they fit and what is open to them. The roadmap page filters it all by phase, by collaborator lane, by domain family and by owner; at zistgah.org/roadmap.

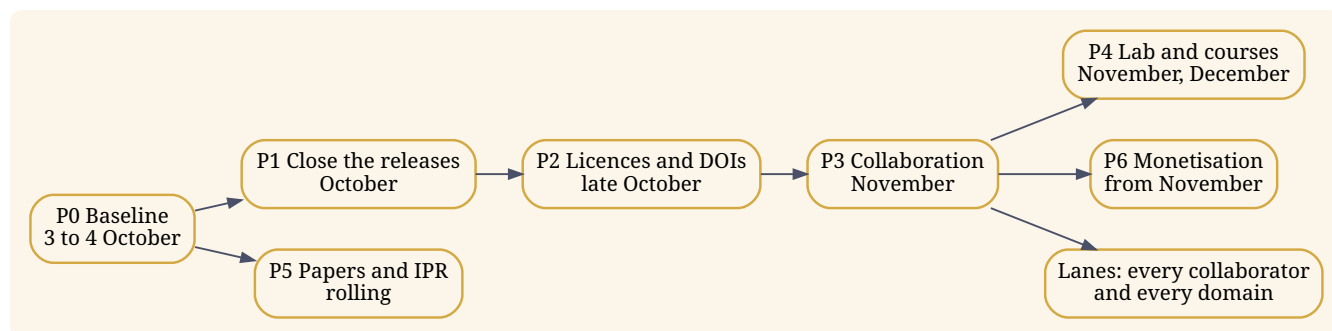


Figure 12. The plan's phases; the lanes run alongside from P3.

Phases

Phase	Window	Goal	Issues
P0 Baseline	Weekend of 3 to 4 October 2026	One measured inventory, the board, tokens with the right scopes, and the list of rulings.	4
P1 Close the releases	10 to 25 October 2026	Everything built is released, and every defect found by execution is fixed.	19
P2 Licences and DOIs	24 October to 8 November 2026	Every repository carries a licence file and correct metadata; minting covers the estate by family.	5
P3 Collaboration infrastructure	31 October to 15 November 2026	A newcomer can go from Rahnuma to a merged pull request in one sitting.	7
P4 Reference lab and courses	November to December 2026	The reference lab runs from launch scripts, with GATE-aligned modules and a pilot.	5

Phase	Window	Goal	Issues
P5 Papers and IPR	Rolling, from October 2026	Disclosure ledger first, then preprints of the four papers with results.	7
P6 Monetisation	From November 2026	Two offers defined and priced, on the dossier standard.	3
Lanes: every collaborator and every domain	Continuous, from October 2026	Every collaborator type and every domain family has a visible lane on the board, with an entry point, open work and a way to propose more.	62

Collaborator lanes

Lane	Start here	Contribute	Issues
School students, 10 to 17	the Rahnuma view for school students; CHAKRA, PANINI's studio, GENIE and the FAKIR dome	learning paths, translations of plain-words summaries, questions	1
Undergraduates: BE, BTech, BSc, BS	the GATE view; Running every component	good first issues, component fixes, capstones seeded from FAKIR	52
Postgraduates and doctoral researchers	the researcher views; Appendices C and D	theses and papers on the open questions, measured results, reviews	52
Teachers, lecturers and lab builders	the teacher view; the reference lab	course modules, lab sheets, pilots, reviews of plain-words text	9
Software engineers	the engineer view; the compilers chapter	code, tests, gates, tooling, pull requests	22
Hardware, embedded and chip engineers	Below the software; zamin, pratik_core_mvp, jugaad28	SPICE, layouts, firmware, device control, ESP32 work	8
Linguists, translators and language workers	the linguist view; writing systems and Pāṇini	language tables, corpora, transliteration rules, measurements	7
Researchers in AI, quantum and cognition	the researcher view; PANINIQ; PEDLER	models, experiments, validation, papers	7
Researchers in the humanities and social sciences	the humanities view; Appendix C	studies of agency, language, knowledge and community	4
Clinicians, health workers and medical researchers	VIKRAM's health section; CEMb and QNBCI	clinical questions, partnerships, review, health-domain tasks	3

Lane	Start here	Contribute	Issues
Community workers, NGOs and village institutions	GramSheel and Project VIKRAM	local needs, pilots, language and field knowledge	30
Farmers, artisans and local enterprises	FAKIR's domain lanes; VIKRAM's livelihood and agriculture sections	domain knowledge, problems worth solving, field testing	24
Artists, writers, filmmakers and musicians	the maker view; the cyclers	works made with the cyclers, recorded as method	3
Makers, roboticists and manufacturers	PANINIphy, pench, transeg	physical realizations, parts, embodiment	6
Policy makers, administrators and public bodies	Appendices C and E; sovereignty in Chapter 19	policy questions, adoption, public infrastructure	5
Lawyers, IPR and standards professionals	provenance and licences; the disclosure ledger	licence review, IPR clearance, standards work	14
Industry partners and companies	Appendix E's adoption path; the component cards	integration, pilots, paid offers, support	9
Funders, sponsors and investors	this roadmap and its measures	funding of lanes, labs and pilots	4
Institutions: colleges, universities and labs	the reference lab and the GATE modules	pilots, courses, joint research	13
AI agents working under a human	any issue's URL; Chapter 22, how to use any AI	issue work through pull requests, with gate output attached	65

Domain families

From FAKIR's lattice: [zistgah/fakir data/lattice.json](#) (ISIC Rev.4, ISCO-08, ISCED-F 2013), 1,505 nodes.

Family	Name	Domains within	Issues
ISIC A	Agriculture, forestry and fishing	54	1
ISIC B	Mining and quarrying	29	1
ISIC C	Manufacturing	232	4
ISIC D	Electricity, gas, steam and air conditioning supply	7	1
ISIC E	Water supply; sewerage, waste management and remediation activities	18	1
ISIC F	Construction	22	1
ISIC G	Wholesale and retail trade; repair of motor vehicles and motorcycles	66	1

Family	Name	Domains within	Issues
ISIC H	Transportation and storage	36	1
ISIC I	Accommodation and food service activities	15	1
ISIC J	Information and communication	42	22
ISIC K	Financial and insurance activities	31	1
ISIC L	Real estate activities	5	1
ISIC M	Professional, scientific and technical activities	35	11
ISIC N	Administrative and support service activities	51	1
ISIC O	Public administration and defence; compulsory social security	11	1
ISIC P	Education	14	7
ISIC Q	Human health and social work activities	21	1
ISIC R	Arts, entertainment and recreation	19	1
ISIC S	Other service activities	26	1
ISIC T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	8	1
ISIC U	Activities of extraterritorial organizations and bodies	3	1
ISCO 1	Managers	46	1
ISCO 2	Professionals	125	1
ISCO 3	Technicians and Associate Professionals	109	1
ISCO 4	Clerical Support Workers	41	1
ISCO 5	Service and Sales Workers	57	1
ISCO 6	Skilled Agricultural, Forestry and Fishery Workers	30	1

Family	Name	Domains within	Issues
ISCO 7	Craft and Related Trades Workers	85	1
ISCO 8	Plant and Machine Operators, and Assemblers	57	1
ISCO 9	Elementary Occupations	50	1
ISCO 0	Armed Forces Occupations	9	1
ISCED 00	Generic programmes and qualifications	6	1
ISCED 01	Education	5	4
ISCED 02	Arts and humanities	13	5
ISCED 03	Social sciences, journalism and information	8	5
ISCED 04	Business, administration and law	10	7
ISCED 05	Natural sciences, mathematics and statistics	13	4
ISCED 06	Information and Communication Technologies (ICTs)	4	16
ISCED 07	Engineering, manufacturing and construction	15	4
ISCED 08	Agriculture, forestry, fisheries and veterinary	9	1
ISCED 09	Health and welfare	12	1
ISCED 10	Services	14	1

The AGI layers are L0 Hardware Definition, L1 Verification, L2 Synthesis/Compilation, L3 Firmware/Embedded, L4 Systems, L5 Parallel, L6 Distributed/Cloud, L7 AI/Deep Learning, L8 Robotics/Embodiment, L9 AGI/ASI. Every domain node, crossed with every layer and every language in ILM's registry, is a possible seed:

$$|S| = |D| \times |L| \times |\Lambda| = 1,505 \times 10 \times 7,867 = 118,398,350.$$

How a collaborator comes in

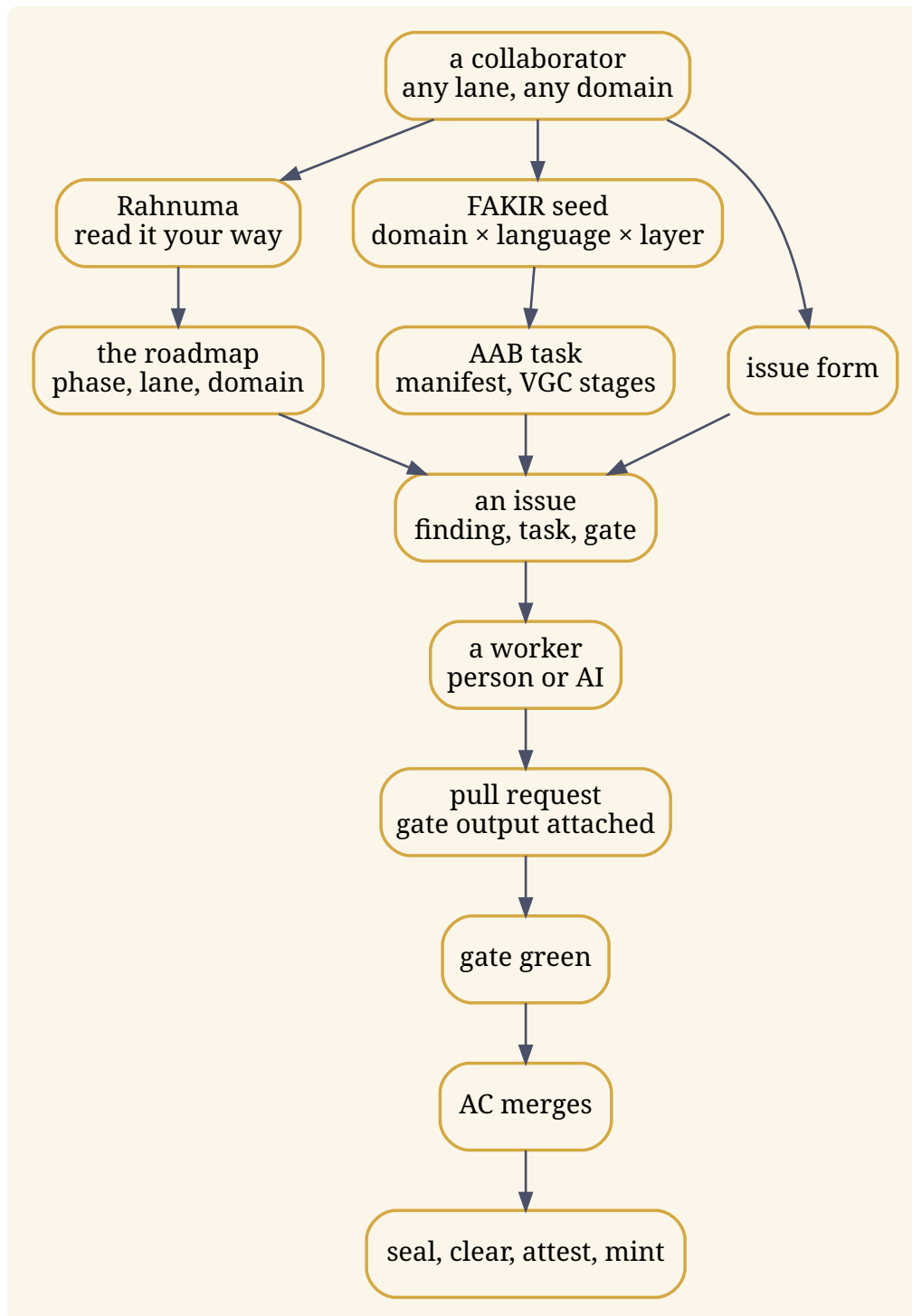


Figure 13. How any collaborator, in any domain, goes from the guide to a sealed contribution.

Four issue forms in [zistgah/governance](#) are the ways in: **Join as a collaborator**, with your lane, domains and languages; **Propose a domain task**, seeded the way FAKIR seeds it, with a domain code, an AGI layer, a language, an intent and a gate; **Report a finding**, with the commands run and what they printed; and **Request a ruling**, for the questions only the author can settle. Every issue says which owner class takes it: a Claude Code session, ChatGPT, Gemini or Fable as workers, or the author himself for push, Pages, mint, merges and rulings.

The work flows the same way for everyone. A worker, a person or an AI, opens a pull request with the gate's output attached; the author merges what is green; anything pushed or minted carries its Candor receipt and goes through seal, clear, attest and mint. Each Saturday the status digest shows every issue as closed, ready, or blocked and by what.

30. What the work would cost: a model-based estimate

In plain words. A measured estimate of how much work the estate holds and what it would cost to rebuild, with and without AI.

How much work does the estate represent, and what would it cost a team to rebuild it, with AI in the process or without? The estimate kit in this guide's repository, under `estimate/`, answers with a recognised model rather than an opinion. It counts every tracked file of every repository once, leaving out vendored, generated, minified and third-party material, measures cyclomatic complexity, and applies COCOMO II.2000, then prices the effort with published salary data and the AI scenario with published token prices and the two controlled trials that bound AI's effect on developer speed. Every rate names its source in `estimate/rates.json`.

Demonstration on 38 repositories cloned for Rahnuma, about a fifth of the estate, most without their full history:

Measure	Value
Unique source lines, each file counted once	243,719
Lines repeated across repositories, counted once	341,227
Words of prose	368,689
Functions measured; above complexity 10; above 20	9,288; 546; 191

Case	Effort, person-months	Schedule, months	Cost, India	Cost, United States
low	701	27.1	₹12.9 crore	\$7.5 million
likely	1,239	35.3	₹41.0 crore	\$18.6 million
high	2,279	47.3	₹130.7 crore	\$49.9 million

AI in the process	Time ratio	Human effort, person-months	AI usage
low	0.44	554	\$140
likely	0.75	944	\$578
high	1.19	1,498	\$5,861

The AI usage costs a few hundred to a few thousand dollars at list prices; the outcome turns on the time ratio, which the trials put between 0.44 and 1.19.

To run it on the whole estate: `bash estimate/estimate.sh --clone`, which clones every repository with full history, or `bash estimate/estimate.sh --root <folder of clones>`. The report also sets the model against the estate as it was actually built, from the distinct days in its git history, with hours per day and AI spend that the author sets in `rates.json`.

COCOMO prices the construction of what is written down. It does not price the research, the invention and the judgement behind it, so every figure here is a floor, not a valuation.

Part VI: Status

31. This release and the next

In plain words. What this edition contains, what changed in it, and what comes next.

This release

Rahnuma 2.0.0 is this guide. It opens with four steps that find each reader's part of it, from a school student to an investor, offers a PDF edition for every kind of reader and every field of study, links school and exam syllabi to the chapters that teach the same ideas on the reader's own device, proposes names for the projections and the procedure for raising a component into Humanesque, publishes the estimate of the work, and marks every object with the author's review. Its policy for versions: when a release changes how the guide is entered or organised, it is a new major version; the previous major version moves, unchanged, to `v<N>/` beside it, and is linked from the cover. [Version 1](#), the edition for advanced readers and senior professionals, stays there as it was. Rahnuma 1.3.0, dated 3 October 2026, was version 1's last release. It adds fourteen diagrams drawn from the recorded relations, an appendix of formal definitions, and the roadmap, with a lane for every collaborator and every domain family (Chapter 29). Version 1.2.0, of the same date, added appendices for researchers in the humanities and social sciences, in science, technology, engineering and mathematics, and for practitioners, an appendix on reading the architecture without flattening it, and the questions behind GramSheel and VIKRAM as the author put them on Gandhi Jayanti. Version 1.1.0, of the same date, supersedes both printings of 1.0.0, one of 53 pages and one of 67, which carried the same version number. It adds the dependency graph of the estate (Chapter 7), the AyeAI Triad, GramSheel and Project VIKRAM, and Humanesque with its projections and their names, each set apart (Chapter 10), the dated entries for GramSheel's beginning in 1993 and VIKRAM's launch in 2020, the entries for Kitab and Research Kundali, and the rule that a record not yet retrieved is not thereby false. It also runs every component of the estate on a clean machine and reports each one as it ran, with the set-up a README leaves out and the reason when a component does not pass (Chapter 15); it gives every chapter a summary in plain words; and on the site it lets a reader choose a background and an age, and shows a reading path, the components to try first, the depth of text, a larger type and a light or dark page. Version 1.0.0 is deposited at [doi:10.5281/zenodo.23062059](https://doi.org/10.5281/zenodo.23062059). Like 1.0.0, it is this guide, in three forms: the site, the PDF, and one Markdown file for reading or for giving to an AI. It comes with its examples, its build and its gate. Every example shown as executed was executed, and `bash ops/verify.sh` runs them again. That includes Hindawi's C shaili, built with Romenagri from its retrieved sources and run through GCC and GDB in Chapter 21.

The estate today

Chapter 12 lists every public repository and every DOI a repository records about itself. Most repositories are pushed and public without a DOI of their own. One case is worth naming because it is often asked about: [zistgah/jyotish](#), the offline panchang, is pushed and its site is live, and it has not been minted. A repository already pushed can be minted later with the estate's seeder; minting is the author's decision, taken repository by repository.

Next

The next release of this package carries the lab launch scripts: one script per lab, the software dockerised and retrieved from the existing repositories, not rebuilt (Chapter 16). The GATE-level courses in computer science and in robotics and automation follow, aligned with the reference lab (Chapter 26).

The consolidated release

This guide marks a plateau: the estate consolidated in one place, with the mathematics written out and the entry points documented. After it, papers from the work are to appear over the coming weeks and months, and the consolidated body over one to two years. In the author's view, ASI and his laboratory fall within that horizon.

Taking part

Open an issue or a pull request on any repository, cite the DOIs, fork freely, and build in your own language.

Colophon

Written and compiled by Abhishek Choudhary, AyeAI, with AI assistance under verification: every example was executed and every statement about the repositories was retrieved from them. ORCID [0009-0002-0684-8320](https://orcid.org/0009-0002-0684-8320). Some address him as Dr; he holds no doctorate. The author is immaterial; the work is what matters.

Typeset in Tiro Devanagari Sanskrit, Noto Serif and its script families, and Noto Sans Mono, all under the SIL Open Font Licence; mathematics rendered by MathJax. Text © 1993–2026 Abhishek Choudhary. All rights reserved. AyeAI. Licensed CC BY-SA 4.0; scripts GPL-3.0-or-later.

Appendices

E. For practitioners: teachers, policy makers, communities, makers and engineers

In plain words. A guide for teachers, policy makers, community workers, artists and engineers: where to start and how to take part.

Teachers and lab builders. Start from the reference lab, the GATE chapter and the component cards, which double as laboratory sheets: each says what to install, what to type and what a student should see (Chapters 16, 26 and 15). The reader views let each student follow a path suited to their age and background, and remember what they have read.

Policy makers and administrators. The questions that matter are in Chapter 10 and Appendix C: sovereignty in its testable technical sense, the ability to rebuild, verify, modify and run (Chapter 21); provenance that makes claims checkable; and infrastructure that leaves capability with communities. The executable contracts of the repositories show what verifiable governance of software looks like in practice.

Community workers and health practitioners. Project VIKRAM's sections cover education, health, livelihood, justice, environment and habitat, food and agriculture, and peace; its health section accepts submissions as issues. The Urdu edition and the Brahmi hub show work in people's own scripts; CHAKRA and Jyotish run offline.

Artists, writers and makers. The cyclers produce print, film, sound, immersive work, embodied work and the record; matba turns posters into a sealed book; GENIE writes and paints; PANINIphy turns a physical idea into parts. Every cycle is recorded as method, so a piece of work can be repeated, shared and credited.

Engineers in industry. Adopt one component at a time: read its card, clone it, run it, read its `CONTRACT.md` and its gate, check its DOI and provenance, then integrate it through its interface. † The notes give the full adoption ladder: the canonical artifact, a one-page explanation, the nearest industry analogue and the exact difference, a working demonstration, the repository, the DOI and timestamp, the interface, a case study, and the commercial route.

Contributors. Fork, open an issue, send a pull request. Every repository's gate runs on your change; every claim must point to its evidence; the AI you use is welcome, and its report that something passed is not the evidence that it did.

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